Math 54 — Week 12

Nonhomogeneous equations: undetermined coefficients and superposition (Nagle 4.4–4.5)

§1 Facts used from Weeks 1–11

Throughout, $a \neq 0$, b, c are real constants, $L[y] = ay'' + by' + cy$, and $p(r) = ar^2 + br + c$ is the auxiliary polynomial.

(W1) **Homogeneous equations** (Week 11, Nagle 4.2–4.3). L is linear: $L[c_1y_1 + c_2y_2] = c_1L[y_1] + c_2L[y_2]$. $L[e^{rt}] = p(r)e^{rt}$. The problem $L[y] = 0$, $y(t_0) = Y_0$, $y'(t_0) = Y_1$ has exactly one solution (Nagle Thm. 1). Let y_1, y_2 solve $L[y] = 0$, with Wronskian $W = y_1y_2' - y_1'y_2$. If $W(\tau) = 0$ at one point τ, then y_1, y_2 are linearly dependent (Nagle Lemma 1). So two independent solutions have $W(t) \neq 0$ at every t, and then every solution is $c_1y_1 + c_2y_2$ (Nagle Thm. 2).

(W2) **The three root cases** (Week 11). Distinct real roots r_1, r_2: e^{r_1t}, e^{r_2t}. A double root r: e^{rt}, te^{rt}, and then $2ar + b = 0$, that is, $p'(r) = 0$. Complex roots $\alpha \pm i\beta$ ($\beta \neq 0$): $e^{\alpha t}\cos\beta t$, $e^{\alpha t}\sin\beta t$.

(W3) **Complex exponentials** (Week 11). $e^{i\theta} = \cos\theta + i\sin\theta$; $\frac{d}{dt}e^{rt} = re^{rt}$ also for complex r. If $z = u + iv$ is complex-valued and a, b, c are real, then $L[z] = L[u] + iL[v]$ with $L[u]$, $L[v]$ real (the computation in Nagle Lemma 2), so $L[\operatorname{Re} z] = \operatorname{Re} L[z]$ and $L[\operatorname{Im} z] = \operatorname{Im} L[z]$.

(W4) **Triangular matrices** (Weeks 4–5). The determinant of a triangular matrix is the product of its diagonal entries, so a triangular matrix with nonzero diagonal is invertible.

(W5) **Solutions of $A\mathbf{x} = \mathbf{b}$** (Week 2, Lay 1.5 Thm. 6). If $A\mathbf{p} = \mathbf{b}$, the solutions of $A\mathbf{x} = \mathbf{b}$ are exactly $\mathbf{p} + \mathbf{v}_h$ with $A\mathbf{v}_h = \mathbf{0}$.

(W6) **The forced mass–spring model** (Nagle 4.1, cited). A mass m on a spring of stiffness k with damping b, pushed by an outside force $F(t)$, has position y with $my'' + by' + ky = F(t)$.

§2 This week

§2.1 The method of undetermined coefficients (Nagle 4.4)

The question. Find one function y with $y'' + 3y' + 2y = 3t$. The guess $y = At$ looks natural, since the right side is a multiple of t. Why does it fail, and what guess works?

An equation $L[y] = f(t)$ with $f \neq 0$ is **nonhomogeneous**; one solution of it is a **particular solution** y_p. This section finds y_p when f is a polynomial, an exponential, a sine or cosine, or a product of these. The idea: choose a family of functions that L maps into itself and that contains f, write y_p in that family with unknown coefficients, and match coefficients.

Example 1 (Nagle): a polynomial right side. For $y'' + 3y' + 2y = 3t$, try $y = At$: then $y' = A$, $y'' = 0$, and $L[y] = 3A + 2At$. Matching $3t$ needs $2A = 3$ and $3A = 0$, which is impossible. The derivative of At produced a constant term that the guess cannot cancel. So add a constant: $y = At + B$ gives

$$y'' + 3y' + 2y = 3A + 2(At + B) = 2At + (3A + 2B).$$

Matching $3t$: $2A = 3$ and $3A + 2B = 0$, so $A = \frac{3}{2}$, $B = -\frac{9}{4}$, and $y_p = \frac{3}{2}t - \frac{9}{4}$.

The question answered. At fails because L sends t to $2t+3$: derivatives lower the degree, so lower powers appear. The guess must contain every power up to the degree of f: $y_p = At + B$, giving $y_p = \frac{3}{2}t - \frac{9}{4}$.

Examples 2 and 3 (Nagle): exponential and sine. **(2)** $y'' + 3y' + 2y = 10e^{3t}$. Try Ae^{3t}: by (W1),

$$L[Ae^{3t}] = Ap(3)e^{3t} = A(9+9+2)e^{3t} = 20Ae^{3t},$$

so $A = \frac{1}{2}$ and $y_p = \frac{1}{2}e^{3t}$.

(3) $y'' + 3y' + 2y = \sin t$. The guess $A\sin t$ fails, since its derivative is a cosine: $L[A\sin t] = A\sin t + 3A\cos t$. Include both: with $y = A\sin t + B\cos t$, $y' = A\cos t - B\sin t$, $y'' = -A\sin t - B\cos t$,

$$y'' + 3y' + 2y = (A - 3B)\sin t + (3A + B)\cos t.$$

Matching $\sin t$: $A - 3B = 1$, $3A + B = 0$, so $A = \frac{1}{10}$, $B = -\frac{3}{10}$, and $y_p = \frac{1}{10}\sin t - \frac{3}{10}\cos t$.

Physically, a damped spring (W6) shaken by $\sin t$ ends up moving at the same frequency (Nagle 4.1, cited), so a combination of $\cos t$ and $\sin t$ is the expected response.

Example 4 (Nagle): polynomial times exponential. $y'' + 4y = 5t^2e^t$. Try $y = (At^2 + Bt + C)e^t$. Then

$$y' = (At^2 + (2A+B)t + B + C)e^t, \qquad y'' = (At^2 + (4A+B)t + 2A + 2B + C)e^t,$$

$$y'' + 4y = \big(5At^2 + (4A+5B)t + (2A+2B+5C)\big)e^t.$$

Matching $5t^2e^t$: $5A = 5$, $4A + 5B = 0$, $2A + 2B + 5C = 0$, so $A = 1$, $B = -\frac{4}{5}$, $C = -\frac{2}{25}$, and

$$y_p = \Big(t^2 - \frac{4}{5}t - \frac{2}{25}\Big)e^t.$$

When the basic guess fails. For $y'' + y' = 5$, the guess $y = A$ gives $L[A] = 0$: the constant solves the homogeneous equation, so no choice of A gives 5. Multiplying by t works: $L[At] = 0 + A = A$, so $A = 5$ and $y_p = 5t$. For $y'' - 6y' + 9y = e^{3t}$, both Ae^{3t} and Ate^{3t} solve the homogeneous equation (3 is a double root), and one needs At^2e^{3t}:

$$L[At^2e^{3t}] = A(2 + 12t + 9t^2)e^{3t} - 6A(2t + 3t^2)e^{3t} + 9At^2e^{3t} = 2Ae^{3t},$$

so $A = \frac{1}{2}$ and $y_p = \frac{1}{2}t^2e^{3t}$. The number of extra factors of t is the number of times the exponent is a root of p. The next formula shows why.

Proposition (The key formula)

For a polynomial $q(t)$ and a real or complex number r,

$$L\big[q(t)e^{rt}\big] = \big(a\,q'' + p'(r)\,q' + p(r)\,q\big)e^{rt}, \qquad p'(r) = 2ar + b.$$

Proof. By the product rule (and (W3) for complex r), $(qe^{rt})' = (q' + rq)e^{rt}$ and $(qe^{rt})'' = (q'' + 2rq' + r^2q)e^{rt}$. So

$$L[qe^{rt}] = \big(a(q'' + 2rq' + r^2q) + b(q' + rq) + cq\big)e^{rt} = \big(aq'' + (2ar+b)q' + (ar^2 + br + c)q\big)e^{rt}. \qquad \square$$

So matching $L[qe^{rt}] = Ct^me^{rt}$ means solving $aq'' + p'(r)q' + p(r)q = Ct^m$ for a polynomial q. Which coefficient leads depends on whether $p(r)$ and $p'(r)$ are 0.

Where the rule comes from (Nagle). Write $q = A_nt^n + A_{n-1}t^{n-1} + A_{n-2}t^{n-2} + \dots$ with $A_n \neq 0$. By the key formula, the three highest powers in $L[qe^{rt}]$ are

$$\begin{aligned}
&A_n\,p(r)\,t^ne^{rt},\\
&\big(nA_n\,p'(r) + A_{n-1}\,p(r)\big)t^{n-1}e^{rt},\\
&\big(n(n-1)A_n\,a + (n-1)A_{n-1}\,p'(r) + A_{n-2}\,p(r)\big)t^{n-2}e^{rt}.
\end{aligned}$$

To match Ct^me^{rt}, the highest power present must be t^m. If $p(r) \neq 0$, the first line is the highest, so $n = m$. If $p(r) = 0 \neq p'(r)$, the first line is 0 and the second is the highest, so $n = m + 1$. If $p(r) = p'(r) = 0$, only the third

remains, so $n = m+2$. In the last two cases the lowest terms $A_0 e^{rt}$ (and $A_1 te^{rt}$) solve the homogeneous equation and can be set to 0, which leaves a factor t (or t^2). The case $f = Ct^m$ is $r = 0$: then one factor t is needed if $c = 0 \neq b$, and two if $b = c = 0$.

Theorem (Method of undetermined coefficients (Nagle 4.4))

(a) For $L[y] = Ct^m e^{rt}$ (r real, $m \geq 0$ an integer) there is a particular solution

$$y_p = t^s(A_m t^m + \cdots + A_1 t + A_0)e^{rt},$$

where $s = 0$ if r is not a root of p, $s = 1$ if r is a **simple root** ($p(t) = a(t-r)(t-r_2)$ with $r_2 \neq r$), and $s = 2$ if r is a double root.

(b) For $L[y] = Ct^m e^{\alpha t}\cos\beta t$ or $Ct^m e^{\alpha t}\sin\beta t$ with $\beta \neq 0$ there is a particular solution

$$y_p = t^s(A_m t^m + \cdots + A_0)e^{\alpha t}\cos\beta t + t^s(B_m t^m + \cdots + B_0)e^{\alpha t}\sin\beta t,$$

where $s = 0$ if $\alpha + i\beta$ is not a root of p and $s = 1$ if it is.

The idea

By the key formula, the problem is to find a polynomial q with $aq'' + p'(r)q' + p(r)q = Ct^m$. If $p(r) \neq 0$, the map $q \mapsto aq'' + p'(r)q' + p(r)q$ keeps the degree of q and is invertible on polynomials of degree $\leq m$, because its matrix is triangular with $p(r)$ on the diagonal. If $p(r) = 0$, the term $p(r)q$ is 0 and the left side has degree one less than q, so q needs degree $m+1$; its constant term solves the homogeneous equation and can be dropped, which leaves a factor t. At a double root $p'(r)q'$ is 0 too, which requires a second factor t.

Proof. Write $\mathbb{P}_m$ for the polynomials of degree at most m, with basis $1, t, \ldots, t^m$, and $M(q) = aq'' + p'(r)q' + p(r)q$.

1. **r is not a root ($s = 0$).** $M(t^k) = p(r)t^k + (\text{lower powers})$, because q' and q'' have lower degree. So M maps $\mathbb{P}_m$ to $\mathbb{P}_m$, and its matrix in the basis $1, t, \ldots, t^m$ is upper triangular with every diagonal entry $p(r) \neq 0$. By (W4) it is invertible, so some $q \in \mathbb{P}_m$ has $M(q) = Ct^m$.

2. **r is a simple root ($s = 1$).** Now $p(r) = 0$, and $p'(r) \neq 0$: from $p(t) = a(t-r)(t-r_2)$, $p'(r) = a(r-r_2) \neq 0$. So

   $$M(q) = aq'' + p'(r)q' = N(q'), \qquad N(u) = au' + p'(r)u.$$

   As in step 1, N is invertible on $\mathbb{P}_m$ (diagonal entries $p'(r) \neq 0$), so some $u \in \mathbb{P}_m$ has $N(u) = Ct^m$. Take $q(t) = \int_0^t u$: then $q' = u$, $M(q) = Ct^m$, and $q(0) = 0$, so $q = t \cdot (\text{polynomial of degree } m)$.

3. **r is a double root ($s = 2$).** Now $p(r) = p'(r) = 0$ (W2), so $M(q) = aq''$, and the polynomial below has $aq'' = Ct^m$:

   $$q = \frac{C}{a(m+1)(m+2)}\, t^{m+2}.$$

   This q is t^2 times a polynomial of degree m.

4. **Part (b): complex exponent $r = \alpha + i\beta$.** The key formula holds for complex r, and M now has complex coefficients. Steps 1 and 2 used only that a triangular matrix with nonzero diagonal is invertible (W4), whose proof is the same for complex entries. If $p(r) \neq 0$, step 1 applies with diagonal entries $p(r)$. If $p(r) = 0$, then r is a simple root: the roots $\alpha \pm i\beta$ differ because $\beta \neq 0$, and since their sum 2α equals $-b/a$, we get $b = -2a\alpha$ and $p'(r) = 2a(\alpha + i\beta) + b = 2ai\beta \neq 0$; so step 2 applies with diagonal entries $p'(r)$. Either way $s \leq 1$, and there is a complex polynomial $Q = Q_1 + iQ_2$ (Q_1, Q_2 real, of degree at most m) with

   $$L\left[t^s Q e^{rt}\right] = Ct^m e^{rt}.$$

5. Since $Ct^m e^{rt} = Ct^m e^{\alpha t}(\cos\beta t + i\sin\beta t)$, taking real and imaginary parts with (W3) gives solutions for the cosine and the sine right sides:

   $$\mathrm{Re}\left(t^s Q e^{rt}\right) = t^s e^{\alpha t}(Q_1\cos\beta t - Q_2\sin\beta t), \qquad \mathrm{Im}\left(t^s Q e^{rt}\right) = t^s e^{\alpha t}(Q_2\cos\beta t + Q_1\sin\beta t).$$

   Both have the form in (b).

□

First-order equations $by' + cy = f$ $(b \neq 0)$. Set $a = 0$: the key formula becomes $L[qe^{rt}] = \big(bq' + p(r)q\big)e^{rt}$ with $p(r) = br + c$ and $p'(r) = b \neq 0$. Steps 1 and 2 of the proof apply, so $s = 0$ unless $r = -c/b$, and then $s = 1$. Instance: for $y' + 2y = 4t$, $y = At + B$ gives $2At + (A + 2B)$, so $A = 2$, $B = -1$ and $y_p = 2t - 1$.

Examples 5 and 6 (Nagle): writing down the form. For $y'' + 2y' - 3y = f$, $p(r) = (r-1)(r+3)$. For $y'' - 2y' + y = f$, $p(r) = (r-1)^2$.

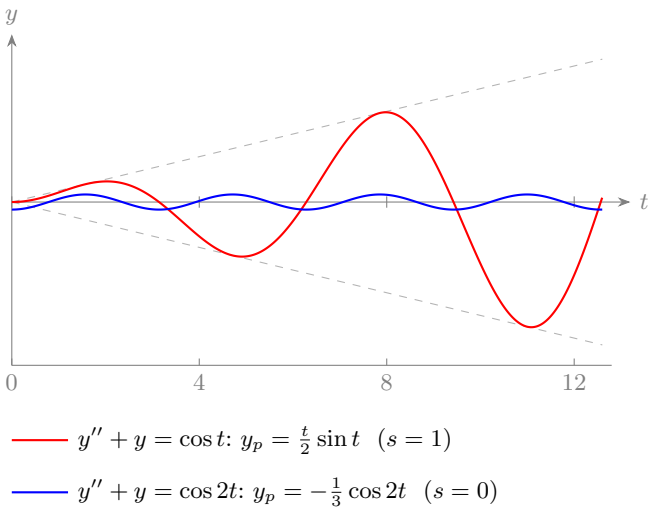
f	$y'' + 2y' - 3y = f$	$y'' - 2y' + y = f$
$7\cos 3t$	$A\cos 3t + B\sin 3t$	same
$2te^t \sin t$	$(A_1t + A_0)e^t\cos t + (B_1t + B_0)e^t \sin t$	same
$t^2 \cos \pi t$	$(A_2t^2 + A_1t + A_0)\cos \pi t + (B_2t^2 + B_1t + B_0)\sin \pi t$	same
$5e^{-3t}$	Ate^{-3t} $\quad(-3$ simple root$)$	Ae^{-3t}
$3te^t$	$t(A_1t + A_0)e^t$ $\quad(1$ simple$)$	$t^2(A_1t + A_0)e^t$ $\quad(1$ double$)$
t^2e^t	$t(A_2t^2 + A_1t + A_0)e^t$	$t^2(A_2t^2 + A_1t + A_0)e^t$

Example 7 (Nagle): a complex root. For $y'' - 2y' + 2y = 5te^t \cos t$, $p(r) = r^2 - 2r + 2$ has roots $1 \pm i$, and $\alpha + i\beta = 1 + i$ is one of them, so $s = 1$:

$$y_p = t(A_1t + A_0)e^t \cos t + t(B_1t + B_0)e^t \sin t.$$

Resonance (Nagle 4.10). When the forcing has the same frequency as a homogeneous solution, $s = 1$ and the amplitude of y_p grows without bound. For $y'' + y = \cos t$, $p(r) = r^2 + 1$ has the root i, and $p'(i) = 2i \neq 0$. The key formula with $q = t$ and $r = i$ gives $L[te^{it}] = p'(i)e^{it} = 2ie^{it}$, so $z = te^{it}/(2i)$ solves $L[z] = e^{it}$, and by (W3) $y_p = \operatorname{Re} z$:

$$z = \frac{t(\cos t + i \sin t)}{2i} = \frac{t}{2}\sin t - \frac{it}{2}\cos t, \qquad y_p = \operatorname{Re} z = \frac{t}{2}\sin t.$$



$p(r) = r^2 + 1$ has roots $\pm i$. For $\cos t$, i is a root, so $s = 1$ and the particular solution grows inside the dashed lines $\pm t/2$ (resonance: the forcing has the frequency of the homogeneous solutions $\cos t$, $\sin t$). For $\cos 2t$, $2i$ is not a root: $s = 0$ and the response stays bounded.

The method needs right sides whose derivatives stay in a finite family. For $\tan t$ (or $\sec t$, $1/t$, $t^{-1}e^t$) the derivatives keep producing new functions, so no finite guess works; Nagle 4.6 treats these by another method. It also needs constant coefficients: $ty'' - y' + 2y = \sin 3t$ is not covered.

Shortcuts

- **Exponential right side**: if $p(r) \neq 0$, then $y_p = \big(C/p(r)\big)e^{rt}$ for $f = Ce^{rt}$. Reason: $L[e^{rt}] = p(r)e^{rt}$ (W1). Instance: Example 2, $y_p = \frac{10}{20}e^{3t}$.

- **Sine or cosine by complex exponential**: for $f = C\sin\beta t$, solve $L[z] = Ce^{i\beta t}$ by $z = \big(C/p(i\beta)\big)e^{i\beta t}$ and take $y_p = \operatorname{Im} z$ (for cos, $\operatorname{Re} z$). Reason: (W3). Condition: $p(i\beta) \neq 0$. Instance: Example 3, $p(i) = 1 + 3i$, and
$$z = \frac{e^{it}}{1+3i} = \frac{(1-3i)(\cos t + i\sin t)}{10}, \qquad \operatorname{Im} z = \frac{\sin t - 3\cos t}{10}.$$
- **Resonant sine or cosine**: if $p(i\beta) = 0 \neq p'(i\beta)$, then $z = \big(C/p'(i\beta)\big)te^{i\beta t}$ solves $L[z] = Ce^{i\beta t}$; take $\operatorname{Re} z$ or $\operatorname{Im} z$. Reason: the key formula with $q = t$ gives $L[te^{rt}] = p'(r)e^{rt}$ for complex r too. Instance: $y'' + y = \cos t$ gives $y_p = \frac{t}{2}\sin t$ (computed above).
- **Finding** s: compute the roots of p once; then s is how many times the exponent (r, or $\alpha + i\beta$) appears among them. Reason: the Theorem. Instance: the table above.
- **Double root shortcut**: if r is a double root, $L[qe^{rt}] = aq''e^{rt}$, so integrate twice. Reason: the key formula with $p(r) = p'(r) = 0$. Instances: $y'' - 6y' + 9y = e^{3t}$ gives $q'' = 1$, $q = \frac{1}{2}t^2$; and $y'' = 2$ (0 is a double root of r^2) gives $q = t^2$, which is two factors of t times the homogeneous solution 1.
- **Exponent is a simple root**: if $p(r) = 0 \neq p'(r)$, then $y_p = \big(C/p'(r)\big)te^{rt}$ for $f = Ce^{rt}$. Reason: the key formula with $q = t$ gives $L[te^{rt}] = p'(r)e^{rt}$. Instance: $y'' + y' = 5$: $r = 0$, $p'(0) = 1$, $y_p = 5t$.

Traps

- Missing lower powers: for $y'' + 3y' + 2y = 3t$ the guess At fails (Example 1); keep all of $t^m, \ldots, t, 1$.
- Missing the partner trig function: for $\sin t$ the guess $A\sin t$ fails (Example 3); always include both $\cos\beta t$ and $\sin\beta t$.
- Forgetting t^s: for $y'' + y' = 5$, the guess A gives $0 = 5$. Check the roots of p before guessing.
- Too few factors of t fail, and so do too many. For $y'' + 3y' + 2y = 10e^{3t}$, the guess Ate^{3t} gives $L[Ate^{3t}] = A(20t + 9)e^{3t}$, which never equals $10e^{3t}$. Use exactly t^s. A complex root is always simple, so $s \leq 1$ in part (b) of the method of undetermined coefficients: never t^2 with cos and sin for a second-order equation.

§2.2 Superposition and the general solution (Nagle 4.5)

The question. We know $y_1 = \frac{3}{2}t - \frac{9}{4}$ solves $y'' + 3y' + 2y = 3t$ and $y_2 = \frac{1}{2}e^{3t}$ solves $y'' + 3y' + 2y = 10e^{3t}$. Find a solution of $y'' + 3y' + 2y = -9t + 20e^{3t}$ without new computations, and then all solutions.

Theorem (Superposition principle (Nagle Thm. 3))
If $L[y_1] = f_1$ and $L[y_2] = f_2$, then $L[k_1y_1 + k_2y_2] = k_1f_1 + k_2f_2$ for all constants k_1, k_2.

Proof. L is linear (W1), so
$$L[k_1y_1 + k_2y_2] = k_1L[y_1] + k_2L[y_2] = k_1f_1 + k_2f_2. \qquad \square$$

Instance (Nagle 4.5, Example 1): $y_1 + y_2 = \frac{3}{2}t - \frac{9}{4} + \frac{1}{2}e^{3t}$ solves $y'' + 3y' + 2y = 3t + 10e^{3t}$.

The question answered (first part). $-9t + 20e^{3t} = -3(3t) + 2(10e^{3t})$, so $-3y_1 + 2y_2 = -\frac{9}{2}t + \frac{27}{4} + e^{3t}$ is a solution.

Theorem (Existence and uniqueness, nonhomogeneous case (Nagle Thm. 4))

Let y_p solve $L[y] = f$ on an interval I containing t_0, and let y_1, y_2 be linearly independent solutions of $L[y] = 0$. Then for all Y_0, Y_1 the problem

$$L[y] = f, \qquad y(t_0) = Y_0, \quad y'(t_0) = Y_1$$

has exactly one solution on I, and it is $y = y_p + c_1 y_1 + c_2 y_2$ for suitable constants c_1, c_2.

The idea

The difference of two solutions of $L[y] = f$ solves $L[y] = 0$, exactly as the difference of two solutions of $A\mathbf{x} = \mathbf{b}$ solves $A\mathbf{x} = \mathbf{0}$ (W5). So all the freedom is in the homogeneous part, whose two constants are fixed by two conditions; the Wronskian is the determinant of that 2×2 system.

Proof. 1. **It is a solution.** By superposition, $L[y_p + c_1 y_1 + c_2 y_2] = f + 0 + 0 = f$.

2. **The conditions can be met.** They read, as a linear system in c_1, c_2,

$$c_1 y_1(t_0) + c_2 y_2(t_0) = Y_0 - y_p(t_0), \qquad c_1 y_1'(t_0) + c_2 y_2'(t_0) = Y_1 - y_p'(t_0).$$

Its determinant is the Wronskian $W(t_0)$, which is not 0 because y_1, y_2 are independent solutions (W1). So it has exactly one solution (c_1, c_2). To find it, write $a = Y_0 - y_p(t_0)$ and $b = Y_1 - y_p'(t_0)$. Multiply the first equation by $y_2'(t_0)$, the second by $y_2(t_0)$, and subtract: the c_2 terms cancel and $c_1 W(t_0) = a y_2'(t_0) - b y_2(t_0)$. Multiply the second by $y_1(t_0)$, the first by $y_1'(t_0)$, and subtract: $c_2 W(t_0) = b y_1(t_0) - a y_1'(t_0)$. Dividing by $W(t_0) \neq 0$,

$$c_1 = \frac{(Y_0 - y_p(t_0))y_2'(t_0) - (Y_1 - y_p'(t_0))y_2(t_0)}{W(t_0)}, \qquad c_2 = \frac{(Y_1 - y_p'(t_0))y_1(t_0) - (Y_0 - y_p(t_0))y_1'(t_0)}{W(t_0)}.$$

3. **Uniqueness.** If $\tilde{y}$ is another solution of the problem, $u = y - \tilde{y}$ has $L[u] = f - f = 0$ and $u(t_0) = u'(t_0) = 0$. The zero function also solves this, so by Nagle Thm. 1 (W1), $u = 0$.

□

So $y = y_p + c_1 y_1 + c_2 y_2$ is the **general solution** of $L[y] = f$: any solution y_g equals it, by choosing c_1, c_2 to match $y_g(t_0)$, $y_g'(t_0)$ and using uniqueness.

The question answered (second part). $p(r) = r^2 + 3r + 2 = (r+1)(r+2)$, so all solutions are

$$y = -\frac{9}{2}t + \frac{27}{4} + e^{3t} + c_1 e^{-t} + c_2 e^{-2t}.$$

Solving $L[y] = f$ with initial conditions.

1. Solve $p(r) = 0$; write the homogeneous solution $c_1 y_1 + c_2 y_2$ (W2).
2. Split f into terms of the forms in the theorem; find y_p for each (one guess can cover several terms with the same exponent), and add (superposition).
3. Apply the initial conditions to the *whole* $y = y_p + c_1 y_1 + c_2 y_2$ and solve for c_1, c_2.

Example 2 (Nagle): initial conditions. $y_p = t^2$ solves $y'' - y = 2 - t^2$: indeed $y_p'' - y_p = 2 - t^2$. The roots of $r^2 - 1$ are ± 1, so $y = t^2 + c_1 e^t + c_2 e^{-t}$. Then $y(0) = c_1 + c_2 = 1$ and $y'(0) = 0 + c_1 - c_2 = 0$ give $c_1 = c_2 = \frac{1}{2}$:

$$y = t^2 + \tfrac{1}{2}(e^t + e^{-t}) = t^2 + \cosh t.$$

Example 3 (Nagle): a forced, damped spring. Mass 1, damping 2, stiffness 2, force $5\sin t + 5\cos t$ (W6), with $y(0) = 1$, $y'(0) = 2$:

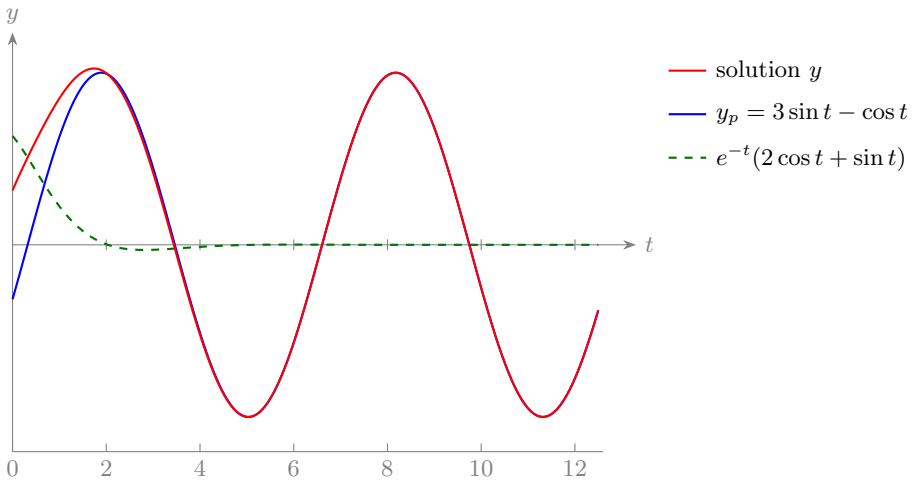
$$y'' + 2y' + 2y = 5\sin t + 5\cos t.$$

The roots of r^2+2r+2 are $-1\pm i$, so the homogeneous solution is $e^{-t}(c_1\cos t+c_2\sin t)$. Since i is not a root, $A\sin t+B\cos t$ is the form for each term. Nagle first solves each one: for $5\sin t$ the guess gives $\sin t-2\cos t$, for $5\cos t$ it gives $2\sin t+\cos t$, and superposition adds them. The same form also handles both terms at once:

$$y_p''+2y_p'+2y_p=(A-2B)\sin t+(2A+B)\cos t,$$

and $A-2B=5$, $2A+B=5$ give $A=3$, $B=-1$. With $y=e^{-t}(c_1\cos t+c_2\sin t)+3\sin t-\cos t$, the conditions are $y(0)=c_1-1=1$ and $y'(0)=-c_1+c_2+3=2$, so $c_1=2$, $c_2=1$:

$$y=2e^{-t}\cos t+e^{-t}\sin t+3\sin t-\cos t.$$



Example 3. The homogeneous part (dashed) decays like e^{-t} (the *transient*); after a few seconds the solution (red) follows the particular solution (blue), which oscillates with the frequency of the force (the *steady state*).

Example 4 (Nagle): one guess for two terms. $y''-y=8te^t+2e^t$. The roots of r^2-1 are ±1, so 1 is a simple root and both terms need $s=1$: $8te^t$ needs $t(A_1t+A_0)e^t$, and $2e^t$ needs tA_0e^t, which is already contained in the first form. With $y_p=(A_1t^2+A_0t)e^t$, the key formula ($q=A_1t^2+A_0t$, $p(1)=0$, $p'(1)=2$, $a=1$) gives

$$L[y_p]=\big(2A_1+2(2A_1t+A_0)\big)e^t=\big(4A_1t+(2A_1+2A_0)\big)e^t.$$

Matching $8te^t+2e^t$: $A_1=2$, $A_0=-1$, so $y_p=(2t^2-t)e^t$.

Theorem (Method of undetermined coefficients, revisited (Nagle 4.5))

For $L[y]=P_m(t)e^{rt}$ with P_m a polynomial of degree m, use $y_p=t^s(A_mt^m+\cdots+A_0)e^{rt}$ with s as before. For $L[y]=P_m(t)e^{\alpha t}\cos\beta t+Q_n(t)e^{\alpha t}\sin\beta t$ ($\beta\neq0$), use

$$y_p=t^s(A_kt^k+\cdots+A_0)e^{\alpha t}\cos\beta t+t^s(B_kt^k+\cdots+B_0)e^{\alpha t}\sin\beta t,\qquad k=\max(m,n),$$

with $s=1$ if $\alpha+i\beta$ is a root of p and $s=0$ otherwise.

Proof. 1. **Exponential case.** Write $P_m=C_mt^m+\cdots+C_0$. Each term $C_jt^je^{rt}$ ($j\le m$) has a particular solution t^s(polynomial of degree j)e^{rt} by part (a) of the method of undetermined coefficients (Nagle 4.4). By superposition their sum solves $L[y]=P_m(t)e^{rt}$, and it is t^s times a polynomial of degree at most m times e^{rt}: the stated form.

2. **Trigonometric case.** Each term $t^je^{\alpha t}\cos\beta t$ or $t^je^{\alpha t}\sin\beta t$ with $j\le k$ has a particular solution of the form in part (b) of the method of undetermined coefficients (Nagle 4.4), with degree j. That form is contained in the form with degree k (set the higher coefficients to 0), so by superposition the sum has the stated form.

□

Examples 5 and 6 (Nagle): forms only. **(5)** $y'' + 2y' + 2y = 5e^{-t}\sin t + 5t^3 e^{-t}\cos t$. The roots are $-1 \pm i$ and $\alpha + i\beta = -1 + i$ is one of them, so $s = 1$, and $k = \max(0, 3) = 3$:

$$y_p = t(A_3t^3 + A_2t^2 + A_1t + A_0)e^{-t}\cos t + t(B_3t^3 + B_2t^2 + B_1t + B_0)e^{-t}\sin t.$$

(6) Higher order works the same way (Nagle Ch. 6, cited: s is the number of times the exponent is a root of p): $y''' + 2y'' + y' = 5e^{-t}\sin t + 3 + 7te^{-t}$ has $p(r) = r(r+1)^2$, with 0 simple and -1 double. Term by term the forms are $Ae^{-t}\cos t + Be^{-t}\sin t$, then Ct, then $t^2(Dt + E)e^{-t}$. (If -1 were a triple root, $7te^{-t}$ would need $t^3(Dt + E)e^{-t}$.) So

$$y_p = Ae^{-t}\cos t + Be^{-t}\sin t + Ct + t^2(Dt + E)e^{-t}.$$

Shortcuts

- **Reuse known particular solutions**: write the new right side as a combination of old ones and combine their solutions (superposition). Instance: the question.
- **One guess for terms with the same exponent**: terms $t^j e^{rt}$ with the same r share one form with the largest power. Reason: the form for $t^j e^{rt}$ with $j \le m$ is contained in the form for $t^m e^{rt}$. Instance: Example 4.
- **Powers of sines and cosines**: rewrite them first, then the method applies. Reason: $\cos 2t = 1 - 2\sin^2 t = 2\cos^2 t - 1$ and $\cos 3t = 4\cos^3 t - 3\cos t$, so $\sin^2 t = \frac{1}{2} - \frac{1}{2}\cos 2t$ and $\cos^3 t = \frac{3}{4}\cos t + \frac{1}{4}\cos 3t$; also $4x\sin^2 x + 4x\cos^2 x = 4x$. Instance: for $y'' + y = \sin^2 t$, the term $\frac{1}{2}$ gives $\frac{1}{2}$, and $-\frac{1}{2}\cos 2t$ gives $\frac{-1/2}{p(2i)}\cos 2t = \frac{1}{6}\cos 2t$, so $y_p = \frac{1}{2} + \frac{1}{6}\cos 2t$.
- **Hyperbolic functions**: $\cosh t = \frac{1}{2}(e^t + e^{-t})$ and $\sinh t = \frac{1}{2}(e^t - e^{-t})$ are two exponential terms. Instance: for $y'' - 4y = \cosh t$, $p(\pm 1) = -3$, so $y_p = -\frac{1}{6}e^t - \frac{1}{6}e^{-t} = -\frac{1}{3}\cosh t$.
- **Check**: substitute y_p back; and $y(t_0)$, $y'(t_0)$ of the final answer must match the data. Instance: in Example 3, $y(0) = 2 - 1 = 1$.

Traps

- Initial conditions go on the whole solution $y_p + c_1y_1 + c_2y_2$, not on the homogeneous part alone. In Example 2, solving $c_1 + c_2 = 1$, $c_1 - c_2 = 0$ happens to agree only because $y_p(0) = y_p'(0) = 0$; in Example 3, ignoring y_p would give $c_1 = 1$, $c_2 = 3$ instead of 2, 1.
- Superposition needs the same left side L. $y_1 = 1$ solves $y'' + y = 1$ and $y_2 = \frac{1}{2}$ solves $y'' + 2y = 1$, but $y_1 + y_2 = \frac{3}{2}$ gives $y'' + y = \frac{3}{2} \neq 2$.
- A product like $t^{-1}e^t$, or $\tan t$, $\sec t$, is not of the allowed form, and neither is an equation with a non-constant coefficient such as $y'' + ty = 1$: for a polynomial q of degree n, $q'' + tq$ has degree $n + 1$, so it never equals 1.

§3 Homework this week

HW11, Nagle Section 4.4: 12, 13, 18, 22, 25, 27, 29, 31 (all on the method of undetermined coefficients, Section 2.1 of this handout). HW11 also includes Lay Chapter 6 Supplementary Exercises 6, 8, 13, 14, 16, 19, 23, 24 (review of Weeks 9–10).

§4 Problem set

P1. Find a particular solution of $y'' - y' - 2y = 4t$.

P2. Find a particular solution of $y'' + 2y' + 5y = 10\cos t$, once with real coefficients and once with the complex exponential.

P3. Find a particular solution of $y'' - 4y = 8e^{2t}$.

P4. Find a particular solution of $y'' + 6y' + 9y = 2e^{-3t}$.

P5. Write the form of a particular solution of $y'' + 4y = f$ (do not solve) for (a) $f = t\sin 2t$; (b) $f = e^{2t}\cos t$; (c) $f = 3t^2e^{2t} + \cos 2t$.

P6. The functions below solve $y'' - y' - 2y = 2t$ and $y'' - y' - 2y = e^t$ respectively:

$$y_1 = -t + \frac{1}{2}, \qquad y_2 = -\frac{1}{2}e^t.$$

Find all solutions of $y'' - y' - 2y = 6t - 4e^t$.

P7. Solve $y'' - y = 2t$ with $y(0) = 0$, $y'(0) = 1$.

P8. Solve $y'' + y = \cos 2t$ with $y(0) = 0$, $y'(0) = 0$, and describe the motion.